BLOG-DOC-07

# Content Model, Article Blocks & Publishing Workflow Specification

Wave-2 owner-authorized contract reconciliation

| | |
|---|---|
| Revision | v1.0-RC2 |
| Date | 11 September 2026 |
| Status | OWNER-AUTHORIZED WAVE-2 RECONCILIATION |
| Supersedes | v1.0-RC1 only where this reconciliation explicitly says so |
| Authority rule | This revision resolves existing contradictions only; it creates no new product scope. |

## Normative reconciliation decisions

### CB-W2-01 Autosave editor instance identity

Autosave identity is Site + Post + Admin account + editor_instance_id. editor_instance_id is a non-empty opaque per-editor-tab identifier, 1-128 characters. Autosave save/latest/recovery operations are scoped to that identity. Two tabs for the same Admin and Post never overwrite one another. A base_post_version mismatch remains a stale-write conflict.

### CB-W2-02 structured_data_inputs

The canonical semantic representation is a JSON object keyed by generated structured-data object type. Each type value is a non-empty object whose fields are JSON scalar values or arrays of JSON scalars. Raw JSON-LD directive keys beginning with @ and arbitrary nested objects remain prohibited. No array-form semantic model remains authoritative.

### CB-W2-03 typed STATIC internal links

INTERNAL destinations continue to distinguish ARTICLE/CATEGORY/TAG/AUTHOR entity targets from STATIC page targets. STATIC targets use static_page_key from the existing frozen static-page registry: ABOUT, CONTACT, PRIVACY, TERMS, DISCLAIMER. STATIC internal links do not use a UUID target_id and do not degrade to arbitrary URL storage.

### CB-W2-06 EMBED provider registry

EMBED blocks remain the existing approved block type. provider_key must resolve through a deployment/configuration-owned approved-provider registry. A provider entry declares source_kind URL or ID; URL providers declare exact allowed HTTPS hosts and ID providers declare an identifier validation pattern. An empty registry is valid and fail-closed. No provider brand is approved by this reconciliation.

## Invariant / count effect

No new ArticleBody type, Post state, public Page ID, permission, service, or product capability is created. Exactly 12 ArticleBody block types and 5 persisted Post states remain authoritative.

## Incorporation rule

Supersedes only the affected clauses in Sections 8-12, 21, 24, 27 and Appendix B of RC1. All unaffected RC1 content remains incorporated unchanged.

## Source inclusion note

The following pages are the complete inherited v1.0-RC1 source document. They remain normative except where a clause is explicitly superseded by the reconciliation pages above. This packaging avoids silently rewriting historical source pages while issuing the controlling v1.0-RC2 contract.

BLOG-DOC-07

# Content Model, Article Blocks & Publishing Workflow Specification

Canonical editorial-content and lifecycle authority for site-owned Articles across the 14-site blogging platform

| 5 | 12 | 20 | 15 |
|---|---|---|---|
| persisted post states | approved body block types | Post capabilities | Article Editor capabilities |

**CORE CONTENT PRINCIPLE**
A Post is one site-owned Article aggregate with explicit structured content, one current lifecycle state, durable version/revision history and private recoverable autosaves. Preview never publishes; autosave never becomes a revision; restoring Trash never silently republishes; every relationship remains inside the trusted SiteContext.

| Document control | Value |
|---|---|
| Status | REVIEW CANDIDATE - not frozen until approved |
| Version | 1.0-RC1 |
| Date | 09 September 2026 |
| Upstream authorities | BLOG-DOC-01 through BLOG-DOC-06 v1.0-RC1 (project-frozen baseline) |
| Primary authority | Post logical content model, article-body schema, lifecycle/workflow semantics, revision/autosave/duplicate/schedule/concurrency/URL-history behavior |
| Change policy | After freeze, changing a persisted lifecycle state, approved block type, required content relationship or transition meaning requires controlled change and traceability updates. |

## Developer lock target

A developer must not invent a new editorial state, treat Preview as Published, turn autosave into revision history, restore Trash directly to public state, accept raw HTML as an article block, copy content across sites, bypass field-level authorization because a field appears inside Post Edit, or silently overwrite a newer post version. This document fixes those semantics; later database/API/page-flow/worker specifications implement them without redefining them.

# Document Map


| Section | Subject |
|---|---|
| 1 | Purpose, authority and document boundary |
| 2 | Frozen upstream invariants and terminology |
| 3 | Canonical content identity and ownership |
| 4 | Post aggregate and logical field model |
| 5 | Field presence, mutability and authorization ownership |
| 6 | Author, taxonomy, media and related-content relationships |
| 7 | SEO and discoverability input fields |
| 8 | Article-body document model |
| 9 | Inline rich-text model |
| 10 | Approved article block registry |
| 11 | Block-by-block payload contracts |
| 12 | Content validation model |
| 13 | Persisted lifecycle states and Preview mode |
| 14 | Canonical publishing state machine |
| 15 | Draft creation and durable editing |
| 16 | Scheduling, rescheduling and unscheduling |
| 17 | Publish and Update Published |
| 18 | Archive behavior |
| 19 | Trash, Restore and Permanent Delete |
| 20 | Duplicate Post semantics |
| 21 | Autosave and recovery |
| 22 | Formal revisions and revision restore |
| 23 | Slug change and URL history |
| 24 | Optimistic concurrency and conflict handling |
| 25 | Bulk operations and per-object lifecycle enforcement |
| 26 | Downstream effects and worker handoff |
| 27 | Isolation, authorization and non-disclosure guardrails |
| 28 | Developer acceptance criteria |
| 29 | Traceability requirements |
| 30 | Dependencies and downstream authority handoff |
| A | Canonical post field matrix |
| B | Article block payload matrix |
| C | Lifecycle transition matrix |
| D | One-page developer execution reference |

**AUTHORITY BOUNDARY**
THIS DOCUMENT OWNS: logical Post/content fields, article-body block semantics, lifecycle status values, Preview semantics, transition legality, schedule semantics, autosave/revision distinction, duplicate behavior, URL-history trigger semantics, optimistic concurrency and publishing eligibility. THIS DOCUMENT DOES NOT OWN: SQL tables/types/indexes/RLS (PDF 08), final URL patterns/slug normalization/redirect routing (PDF 09), exact Admin/public UI flows (PDFs 10/11), public component layout (PDF 12), public response projections (PDF 13), HTTP endpoints (PDF 15), SEO algorithms/schema output (PDF 16), media processing/storage (PDF 17), queue/cache syntax (PDF 18), or security/session implementation (PDF 19).

# 1. Purpose, Authority and Document Boundary

## 1.1 Purpose

BLOG-DOC-07 converts the approved content and publishing capabilities into one deterministic semantic model. It answers what a Post contains, which article blocks are valid, which lifecycle states can exist, which transitions are legal, how drafts/schedules/public versions behave, how autosaves differ from revisions, what Duplicate copies, how URL history is triggered and how stale concurrent edits are rejected.

## 1.2 Authority order

| Priority | Authority | Controls relevant to DOC-07 |
|---|---|---|
| 1 | BLOG-DOC-01 | Product topology, 14-site estate, centralized platform, public-content/publication boundaries. |
| 2 | BLOG-DOC-02 | Exact permissions, site assignments, authorization order, object ownership and composite authorization paths. |
| 3 | BLOG-DOC-03 | Which Post/editor/publishing capabilities exist and whether they are REQUIRED/CONDITIONAL. |
| 4 | BLOG-DOC-04 | site_id, SiteContext, same-site relationship integrity, Preview/worker/isolation semantics. |
| 5 | BLOG-DOC-05 | Public Article/Preview page existence and public eligibility/indexability boundaries. |
| 6 | BLOG-DOC-06 | Admin Post/Create/Edit/Revision page IDs and capability-to-page allocation. |
| 7 | BLOG-DOC-07 | Canonical content model and lifecycle/workflow semantics defined here. |

## 1.3 Conflict rule

**CONFLICT RULE**
If implementation convenience conflicts with a frozen upstream rule, the implementation changes; this document must not broaden product scope. If a later specialist document requires a semantic change to a state, block or transition defined here, DOC-07 must be explicitly reopened rather than silently overridden.

# 2. Frozen Upstream Invariants and Terminology

| Invariant ID | Locked invariant |
|---|---|
| CT-INV-01 | Each Post has exactly one immutable owning site_id. |
| CT-INV-02 | All ordinary editorial operations execute inside trusted CTX-ADMIN-SITE; client site_id is never ownership authority. |
| CT-INV-03 | Published content alone is eligible for ordinary PUB-ARTICLE-01 delivery; Draft/Scheduled/Archived/Trash are not. |
| CT-INV-04 | PUB-SYS-PREVIEW-01 is private/noindex and does not change publication state. |
| CT-INV-05 | Posts, revisions/autosaves, author attribution, taxonomy, media, SEO inputs, URL history and scheduled work remain site-owned. |
| CT-INV-06 | Cross-site Post duplication/syndication is not a v1 capability. |

| Invariant ID | Locked invariant |
|---|---|
| CT-INV-07 | Article body is structured editorial content; no unrestricted page builder or raw frontend-code editor exists. |
| CT-INV-08 | H1 belongs to Post title/page template; body headings are H2/H3/H4. |
| CT-INV-09 | Site assignment grants membership only; action permission still comes from BLOG-DOC-02. |
| CT-INV-10 | Wrong-site target IDs are denied/non-disclosed and never loaded tenant-blind. |

### 2.1 Key terms

| Term | Meaning |
|---|---|
| Post | Canonical site-owned Article aggregate managed by Central Admin. |
| Current committed version | Last durable Post state/content accepted under optimistic concurrency. |
| Working copy | Unsaved or autosaved editor content not yet committed to the canonical Post. |
| Preview | Private render of an eligible committed/working copy through the owning public frontend; not a status. |
| Revision | Immutable durable historical snapshot created by an explicit committed mutation; autosave is not a revision. |
| Autosave | Private recoverable editing snapshot tied to a base version; does not publish or create formal history. |
| Schedule | Current future publication instruction for a Post in SCHEDULED status. |
| URL history | Site/post-owned record of prior public locator identity created when an already-publicly-addressable Post changes locator input. |

## 3. Canonical Content Identity and Ownership

A Post is a site-owned aggregate. Its content fields and related references may change, but its post_id and site_id do not change through ordinary editor actions, revision restore, duplicate source mutation, archive, trash or restore. Duplicate creates a new Post identity in the same SiteContext; it does not move the source.

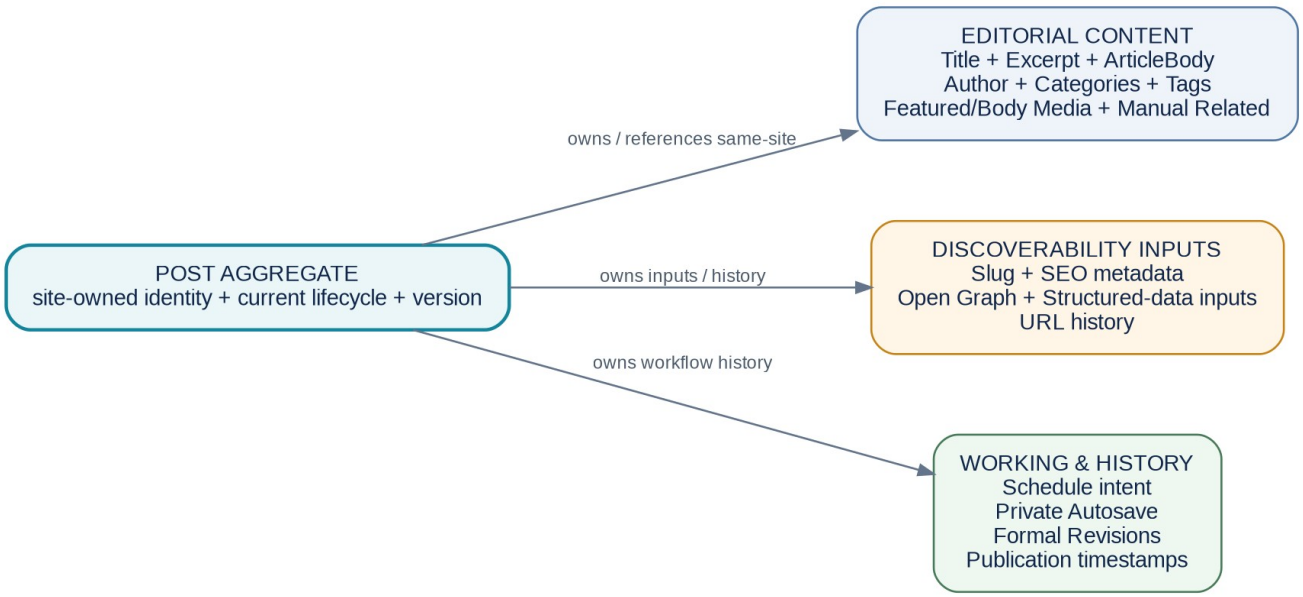



*Figure 1 - Logical Post aggregate and owned/related content concerns. Exact persistence is PDF 08 authority.*

### 3.1 Ownership rules

- createPost derives site_id from trusted SiteContext; payload.site_id is not authoritative.
- Every author/taxonomy/media/related-post reference is resolved inside the same site before commit.
- A Post cannot be reassigned to another site. v1 has no move-to-site operation.
- Revision/autosave/URL-history records inherit the Post site_id and may never point to a different-site Post.
- A worker acting on scheduled publication must reload the Post under the persisted site context before transition.

## 4. Post Aggregate and Logical Field Model

The following is the canonical logical contract. Names are semantic identifiers, not SQL column declarations; PDF 08 may choose physical types/normalization while preserving these meanings.

| Field ID | Logical field | Group | Presence | Semantic rule |
|---|---|---|---|---|
| PST-FLD-001 | post_id | System | Required | Immutable stable Post identity. |
| PST-FLD-002 | site_id | Ownership | Required | Immutable owner derived from trusted SiteContext. |
| PST-FLD-003 | content_schema_version | Content | Required | Version of the article-body schema used to validate body. |
| PST-FLD-004 | status | Lifecycle | Required | One of DRAFT, SCHEDULED, PUBLISHED, ARCHIVED, TRASH. |
| PST-FLD-005 | version | Concurrency | Required | Monotonically increasing optimistic-concurrency token for durable mutation. |
| PST-FLD-006 | title | Editorial | May be blank in early DRAFT; required for schedule/publish | Page/article title; owns H1 in presentation once supplied. |
| PST-FLD-007 | excerpt | Editorial | Optional | Editorial summary; absence is not a publish blocker unless later template contract explicitly requires it. |
| PST-FLD-008 | body | Editorial | Required structure | Ordered ArticleBody document; at least one meaningful block for publish/schedule. |
| PST-FLD-009 | author_site_profile_id | Relationship | Optional in early Draft; required for publish/schedule | One same-site author profile; v1 does not add co-authoring. |
| PST-FLD-010 | category_ids | Relationship | 0..N | Unique same-site category references. |
| PST-FLD-011 | tag_ids | Relationship | 0..N | Unique same-site tag references. |
| PST-FLD-012 | featured_media_id | Media | Optional | Same-site media reference used as featured image where presentation consumes it. |
| PST-FLD-013 | manual_related_post_ids | Relationship | 0..N | Unique same-site Post refs; no self-reference; automatic related remains derived/conditional. |
| PST-FLD-014 | slug | Routing/SEO | Required for public eligibility | Current site-scoped slug input; exact normalization/path is PDF 09 authority. |
| PST-FLD-015 | seo_title | SEO | Optional | Explicit search/social title input; fallback algorithm belongs to PDF 16. |
| PST-FLD-016 | meta_description | SEO | Optional | Explicit meta description input. |
| PST-FLD-017 | canonical_override | SEO | Optional | Validated explicit canonical override; legitimacy/algorithm belongs to PDF 16. |
| PST-FLD-018 | index_policy | SEO | Optional/inherit | Approved index/noindex input, not raw robots markup. |
| PST-FLD-019 | link_robots_policy | SEO | Optional/inherit | Approved follow/nofollow input where applicable. |
| PST-FLD-020 | og_title | SEO | Optional | Explicit Open Graph title input. |
| PST-FLD-021 | og_description | SEO | Optional | Explicit Open Graph description input. |
| PST-FLD-022 | og_image_media_id | SEO/Media | Optional | Same-site media reference for approved social image. |

| Field ID | Logical field | Group | Presence | Semantic rule |
|---|---|---|---|---|
| PST-FLD-023 | structured_data_inputs | SEO | Optional structured object | Validated source fields only; raw JSON-LD is prohibited. |
| PST-FLD-024 | breadcrumb_label | SEO/IA | Optional | Approved breadcrumb source label/input; exact breadcrumb generation is PDF 16. |
| PST-FLD-025 | scheduled_for_utc | Publication | Required only in SCHEDULED | Single absolute publication instant. |
| PST-FLD-026 | schedule_local_input | Publication | Required only in SCHEDULED | Editor-entered local date/time retained for traceability. |
| PST-FLD-027 | schedule_timezone_id | Publication | Required only in SCHEDULED | Site timezone used when converting schedule_local_input to instant. |
| PST-FLD-028 | first_published_at | Publication | Nullable until first publish | Immutable first successful transition to PUBLISHED. |
| PST-FLD-029 | published_at | Publication | Nullable unless currently/previously published | Timestamp of latest transition into the current/most recent PUBLISHED episode; Update Published does not reset it. |
| PST-FLD-030 | archived_at | Lifecycle | Set while ARCHIVED | Timestamp of current transition to ARCHIVED; past events remain in revisions/audit. |
| PST-FLD-031 | trashed_at | Lifecycle | Set while TRASH | Timestamp of current transition to TRASH. |
| PST-FLD-032 | created_at | System | Required | Initial durable creation timestamp. |
| PST-FLD-033 | created_by_admin_id | System | Required for Admin create | Stable creating Admin identity reference; exact FK is PDF 08. |
| PST-FLD-034 | updated_at | System | Required | Timestamp of latest durable mutation. |
| PST-FLD-035 | updated_by_admin_id | System | Nullable for system transition | Actor for latest durable mutation; system worker actor represented explicitly when applicable. |

**NO GENERIC PAGES MODEL**
The five fixed static/information page types frozen by BLOG-DOC-05 are not silently converted into Posts or a generic page-builder content type by this document. DOC-07 governs Article/Post content.

## 5. Field Presence, Mutability and Authorization Ownership

Page location does not grant field authority. ADM-S-POST-03 may surface Post-local SEO/media/author/taxonomy controls, but each mutation still follows BLOG-DOC-02. DOC-07 records the field ownership relationship without creating new permissions.

| Field concern | Frozen authorization path | DOC-07 rule |
|---|---|---|
| Core title/excerpt/body | posts.view + posts.edit | Create mode uses posts.view + posts.create until draft exists. |
| Author assignment/reassignment | posts.view + posts.edit + authors.view | authors.edit does not authorize Post attribution. |
| Category assignment | posts.view + posts.edit + categories.view | Category definition edit is separate; all targets same-site. |
| Tag assignment | posts.view + posts.edit + tags.view | Tag definition edit/merge/delete is separate. |
| Featured/existing body media selection | posts.view + posts.edit + media.view | Uploading within the editor additionally requires media.upload. |
| Manual related Posts | posts.view + posts.edit | Candidate/search read occurs only inside same SiteContext. |
| Slug and dedicated Post SEO fields | seo.view + seo.edit | posts.edit alone does not substitute for SEO authority. |
| Open Graph image selection | seo.view + seo.edit + media.view | Uploading a new asset additionally requires media.upload. |
| Post-local image metadata | posts.view + posts.edit | Media Library metadata instead requires media.view + media.edit per DOC-02. |

### 5.1 Mutability by lifecycle

| Status | Editable content? | Durable save behavior | Public effect |
|---|---|---|---|
| DRAFT | Yes | Save Draft commits new version/revision. | None. |
| SCHEDULED | Yes | Edit commit preserves schedule only if schedule remains valid; schedule change is separately authorized. | No public effect until publish time. |
| PUBLISHED | Working edits yes | Update Published explicitly commits; autosave/working copy does not alter current public content. | Successful update replaces current public eligible version and triggers downstream refresh. |
| ARCHIVED | Yes | Edit commit preserves ARCHIVED status. | None until explicit publish/schedule. |
| TRASH | No ordinary content edit | Restore first. Permanent Delete may be available subject to retention. | None. |

## 6. Author, Taxonomy, Media and Related-Content Relationships

| Relationship | Cardinality | Integrity rule |
|---|---|---|
| Author | Exactly 0..1 while early Draft; exactly 1 before schedule/publish | Must resolve to same-site valid author profile. Retired author cannot be newly assigned, but existing historical attribution remains. |
| Categories | 0..N unique refs | All same-site. No category is required for publish by DOC-07. Category hierarchy belongs taxonomy authority. |
| Tags | 0..N unique refs | All same-site. Tag merge may rewrite relationships without changing Post identity/history. |
| Featured media | 0..1 | Same-site media only. Missing featured media is not a publish blocker. |
| Body media | 0..N via MEDIA blocks | Every media_id same-site; post-local caption/alt override does not transfer asset ownership. |
| Manual related Posts | 0..N unique refs | Same-site only, no self-reference; must not include TRASH or wrong-site targets. Public eligibility is resolved when rendering. |

### 6.1 Same-site validation order

1. Resolve target by site_id + target_id inside trusted SiteContext.
2. Verify target exists and is eligible for the relationship operation.
3. Apply exact BLOG-DOC-02 permission path for the field/action.
4. Persist relation only after all site and business rules pass.
5. Never report that a wrong-site UUID exists; fail with the same safe semantics as unavailable/not-authorized target.

## 7. SEO and Discoverability Input Fields

DOC-07 defines which Post-level SEO inputs are part of the content aggregate. It does not define fallback order, canonical construction, robots emission, JSON-LD generation, sitemap inclusion algorithms or redirect precedence; those remain later specialist authorities.

| Input | Presence | Semantic boundary |
|---|---|---|
| Slug | Required before public eligibility | Site-scoped locator input. Mutation requires SEO authorization even when surfaced in Post Edit. |
| SEO title | Optional | Explicit value may override fallback later; absence is advisory, not a publish blocker. |
| Meta description | Optional | Explicit description input; absence can use later fallback. |
| Canonical override | Optional | Must be valid/authorized; exact allowed cases later. |
| Index policy | Optional/inherit | Approved enum input; no raw robots meta string. |

| Input | Presence | Semantic boundary |
|---|---|---|
| Link robots | Optional/inherit | Approved follow/nofollow input only where applicable. |
| OG title/description/image | Optional | Image reference same-site. Missing values can fall back later. |
| Structured data inputs | Optional structured | Typed/validated source data; no raw JSON-LD text editor. |
| Breadcrumb input | Optional | Source label/data only; automatic construction later. |

**SEO WARNINGS ARE ADVISORY**
A missing optional custom SEO title, description, featured/OG image, tag or related Post does not itself block publication. Hard publication blockers are limited to content/state/site/integrity requirements defined in Section 12.

# 8. Article-Body Document Model

The Post body is an ordered, versioned structured document. The storage format may be JSON or another structured representation selected later, but it must preserve the following semantic envelope and must never degrade into unrestricted HTML/source code.

| Logical property | Required rule |
|---|---|
| content_schema_version | Required positive/versioned schema identifier. New block types or breaking payload changes require a content-schema version decision. |
| blocks | Ordered array. Order is editorial meaning; arbitrary frontend layout coordinates are not stored. |
| block_id | Stable unique identifier within one ArticleBody document. Reordering does not regenerate IDs. |
| type | One approved block type from Section 10 only. |
| payload | Type-specific validated data; unknown keys may not silently become visual controls. |
| schema validation | Every committed save, preview, schedule and publish validates block envelope and type payload. |

## 8.1 Presentation boundary

- No block stores arbitrary CSS, class names, inline style, coordinates, responsive breakpoints or frontend component source.
- CTA/callout/disclaimer carry content/semantic intent only; each site frontend controls visual rendering.
- No arbitrary nested page-builder tree is introduced.
- Block serialization must remain portable across all 14 frontend applications while each frontend owns its presentation.

# 9. Inline Rich-Text Model

Inline rich text is intentionally small. The editor supports text content with bold, italic and link marks. It does not provide arbitrary font family, font size, color, alignment, raw HTML, scripts or inline CSS controls.

| Inline construct | Allowed payload | Rules |
|---|---|---|
| TEXT | Unicode text | Sanitized/validated; empty-only structures do not satisfy meaningful-content validation. |
| BOLD | Mark on text range | Semantic emphasis only; site frontend chooses visual treatment. |
| ITALIC | Mark on text range | Semantic emphasis only. |
| LINK / INTERNAL | label + same-site target reference | Target must resolve inside same SiteContext; exact public URL generated later. |
| LINK / EXTERNAL | label + validated destination URL | Security/allowlist/scheme rules later; raw HTML attributes are not editable. |

## 10. Approved Article Block Registry

| Block ID | Type | Capability | Purpose |
|---|---|---|---|
| CB-PARAGRAPH | PARAGRAPH | FC-EDIT-01 | Rich-text paragraph. |
| CB-HEADING | HEADING | FC-EDIT-02 | Semantic H2/H3/H4 body heading; H1 prohibited. |
| CB-LIST | LIST | FC-EDIT-04 | Ordered or unordered list of rich-text items. |
| CB-QUOTE | QUOTE | FC-EDIT-05 | Editorial quotation with optional attribution/source text. |
| CB-SEPARATOR | SEPARATOR | FC-EDIT-06 | Semantic content break with no arbitrary visual styling. |
| CB-MEDIA | MEDIA | FC-EDIT-08 | Same-site media reference with approved post-local usage metadata. |
| CB-TABLE | TABLE | FC-EDIT-09 | Simple rectangular editorial table. |
| CB-EMBED | EMBED | FC-EDIT-10 | Approved provider/type embed; never raw iframe/HTML. |
| CB-CALLOUT | CALLOUT | FC-EDIT-11 | Approved note/callout content; presentation frontend-controlled. |
| CB-CTA | CTA | FC-EDIT-12 | Approved CTA label/supporting text/destination; no visual code. |
| CB-DISCLAIMER | DISCLAIMER | FC-EDIT-13 | Approved disclaimer content block. |
| CB-SOURCES | SOURCES_REFERENCES | FC-EDIT-14 | Ordered source/reference entries. |

FC-EDIT-03 Text Emphasis and FC-EDIT-07 Links are inline capabilities used inside compatible rich-text-bearing blocks. FC-EDIT-15 Editor Validation applies to the complete document rather than defining a separate block.

**NO RAW BLOCK ESCAPE HATCH**
RAW_HTML, SCRIPT, STYLE, arbitrary IFRAME, arbitrary component JSON and unknown custom block types are prohibited. A new block type requires an explicitly versioned schema change; it is not accepted because a frontend happens to know how to render it.

## 11. Block-by-Block Payload Contracts

| Type | Logical payload | Validation / boundary |
|---|---|---|
| PARAGRAPH | content: RichText | content must contain meaningful text or allowed inline content. |
| HEADING | level: 2\|3\|4; content: RichText | level 1 is invalid; heading content must not be empty. |
| LIST | style: ORDERED\|UNORDERED; items: RichText[] | At least one non-empty item; nesting beyond approved editor behavior is not assumed. |
| QUOTE | quote: RichText; attribution?: text; source?: text/url | Quote required; attribution/source optional and not executable markup. |
| SEPARATOR | no author-controlled visual payload | Represents a semantic break only. |
| MEDIA | media_id; alt_override?; caption? | media_id same-site and visible to operator; post-local override does not mutate Media Library metadata. |
| TABLE | header_row: bool; rows: Cell[][] | Rectangular structure; at least 1 row/1 cell; cells use safe text/rich text; no formulas/scripts/nested blocks. |
| EMBED | provider_key; source_url_or_id; caption? | Provider/type must be approved; no raw embed code; invalid provider blocks save/publish. |
| CALLOUT | label?; body: RichText | Body required; label optional. No arbitrary theme/style token stored. |
| CTA | label; supporting_text?; destination | Destination is validated internal same-site reference or external URL; frontend owns button/layout style. |
| DISCLAIMER | label?; body: RichText | Body required; site frontend controls treatment. |
| SOURCES_REFERENCES | entries[]: title + optional url/publisher/note | At least one valid entry; URLs validated; ordering preserved. |

### 11.1 Block identity and editing

- Block IDs are unique within one body document.
- Move/reorder preserves block_id.
- Copy/paste inside a body creates fresh IDs for copied blocks to prevent duplicate identity.
- A revision snapshot preserves block IDs as they existed in that revision.
- Unknown block types fail validation; they are not silently discarded during save or publish.

## 12. Content Validation Model

Validation has hard blockers and advisory warnings. A hard blocker prevents the requested durable lifecycle action. Advisory warnings may inform editors but do not become hidden publishing requirements.

| Validation ID | Concern | Server-authoritative rule |
|---|---|---|
| VAL-CNT-001 | Trusted active site | Post and every relation resolve under one active SiteContext. |
| VAL-CNT-002 | Title | May be blank in early DRAFT; must be non-blank after normalization before Schedule/Publish and then owns public H1. |
| VAL-CNT-003 | Body | Valid ArticleBody schema and at least one meaningful content block for schedule/publish. |
| VAL-CNT-004 | Heading levels | No H1 inside body; HEADING only level 2/3/4. |
| VAL-CNT-005 | Author | One valid same-site author required for schedule/publish. |
| VAL-CNT-006 | Taxonomy | Any supplied category/tag IDs exist within same site; zero categories/tags is allowed. |
| VAL-CNT-007 | Media | Any referenced media exists within same site and is usable for the operation. |
| VAL-CNT-008 | Related Posts | Unique same-site refs, no self-reference; invalid/wrong-site targets rejected. |
| VAL-CNT-009 | Slug | Required for schedule/publish; site-scoped normalization/conflict validation is delegated to PDF 09 but must pass before public eligibility. |
| VAL-CNT-010 | SEO inputs | Any supplied optional values conform to their approved typed/URL/policy constraints; no raw JSON-LD. |
| VAL-CNT-011 | Links | Internal targets same-site; external destinations pass approved validation; no executable markup. |
| VAL-CNT-012 | Embeds | Provider/type approved; raw iframe/HTML prohibited. |
| VAL-CNT-013 | Schedule | Future, unambiguous instant with retained timezone context. |
| VAL-CNT-014 | Concurrency | expected_version equals current version for every durable mutation. |
| VAL-CNT-015 | Lifecycle | Requested transition exists in Section 14 and source status is eligible. |

### 12.1 Advisory-only baseline

- Missing optional excerpt.
- Missing custom SEO title/meta description when fallback exists.
- Missing featured media or custom OG image.
- No categories or tags.
- No manual related Posts.
- SEO warning conditions that do not violate a hard input constraint.

# 13. Persisted Lifecycle States and Preview Mode

| ID | Value | Class | Meaning |
|---|---|---|---|
| PST-ST-DRAFT | DRAFT | Persisted | Private editable working content. Ordinary public Article route must not serve it. |
| PST-ST-SCHEDULED | SCHEDULED | Persisted | Private content with one active future publication instant. |
| PST-ST-PUBLISHED | PUBLISHED | Persisted | Current committed version is eligible for ordinary public Article delivery/discovery subject to later SEO/routing rules. |
| PST-ST-ARCHIVED | ARCHIVED | Persisted | Retained editorial record removed from ordinary public availability. |
| PST-ST-TRASH | TRASH | Persisted | Recoverable soft-delete state. No ordinary edit/preview/public eligibility. |
| PST-MODE-PREVIEW | PREVIEW | Mode, not status | Private render through PUB-SYS-PREVIEW-01; no lifecycle transition and no public cache/search/sitemap eligibility. |
| PST-ACT-RESTORE | RESTORE | Action, not status | Authorized TRASH -> DRAFT transition. |

**WHY RESTORE RETURNS TO DRAFT**
BLOG-DOC-02 gives Restore its own permission but does not implicitly grant Publish or Schedule. Therefore restoring TRASH always returns the Post to private DRAFT. Republishing or scheduling requires its own explicit authorized action, preventing an authorization bypass.

## 13.1 Preview eligibility

- DRAFT, SCHEDULED and ARCHIVED may be previewed when posts.view + posts.preview and Preview SiteContext/grant validation succeed.
- PUBLISHED may use Preview only for a private working/autosaved edit before Update Published; the currently committed public version remains unchanged.
- TRASH is not previewable through normal Preview; Restore is required first.
- Preview response is private/no-store/noindex and does not populate ordinary public caches/search indexes/sitemaps.

# 14. Canonical Publishing State Machine

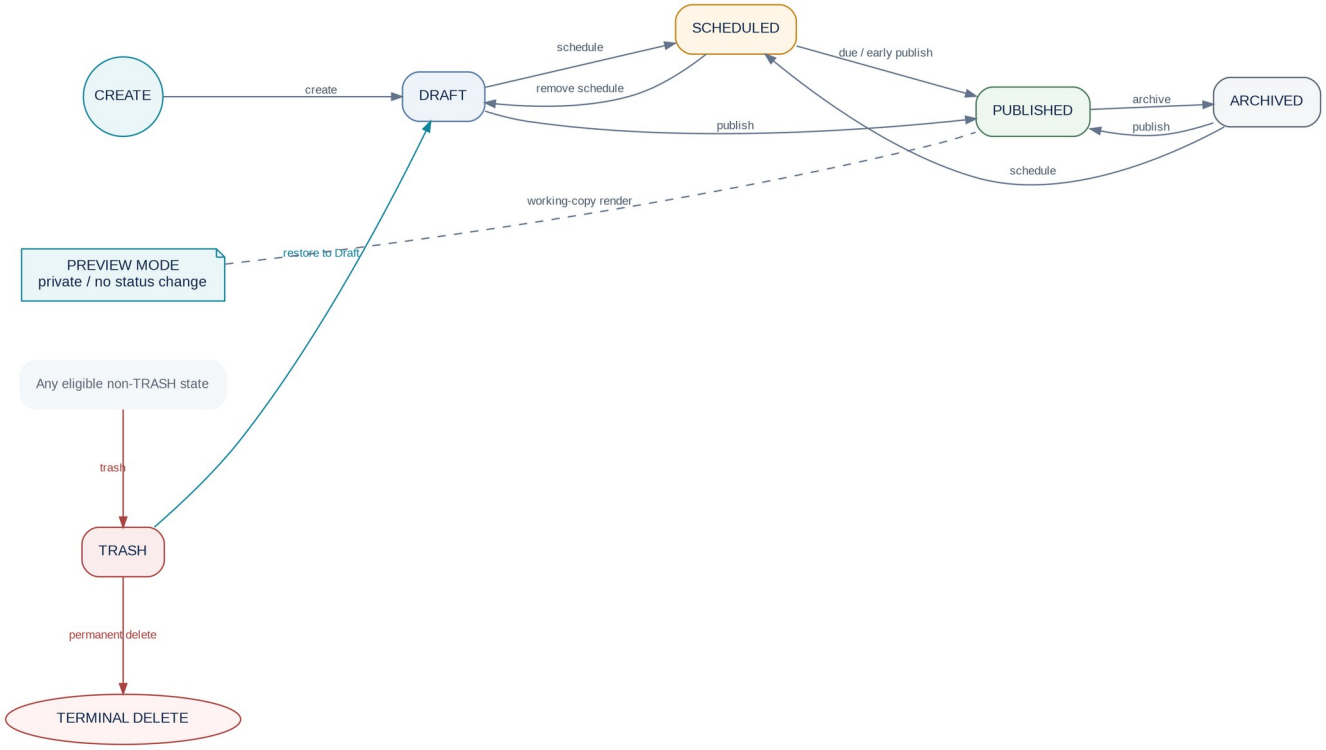



*Figure 2 - Canonical lifecycle. Preview is a side mode; Permanent Delete is terminal.*

| WF ID | Action | From | To | Authorization reference | Semantic result |
|---|---|---|---|---|---|
| WF-001 | Create | NONE | DRAFT | posts.view + posts.create | Create new site-owned draft. |
| WF-002 | Save Draft | DRAFT | DRAFT | posts.view + posts.edit | Commit draft edits; new version/revision. |
| WF-003 | Schedule | DRAFT | SCHEDULED | posts.view + posts.edit + posts.schedule | Validate content + future schedule. |
| WF-004 | Reschedule | SCHEDULED | SCHEDULED | posts.view + posts.edit + posts.schedule | Replace active schedule; stale job becomes invalid. |
| WF-005 | Remove Schedule | SCHEDULED | DRAFT | posts.view + posts.edit + posts.schedule | Clear active schedule; content remains private. |
| WF-006 | Manual Publish | DRAFT | PUBLISHED | posts.view + posts.edit + posts.publish | Publish eligible draft now. |
| WF-007 | Early/Manual Publish | SCHEDULED | PUBLISHED | posts.view + posts.edit + posts.publish | Invalidate active schedule and publish now. |
| WF-008 | Scheduled Publish | SCHEDULED | PUBLISHED | SYSTEM worker under CTX-WORKER-SITE | Worker revalidates site, ownership, schedule identity/version and eligibility. |
| WF-009 | Update Published | PUBLISHED | PUBLISHED | posts.view + posts.edit | Commit working changes to public current version. |
| WF-010 | Archive Published | PUBLISHED | ARCHIVED | posts.view + posts.edit + posts.archive | Remove ordinary public eligibility. |
| WF-011 | Archive Scheduled | SCHEDULED | ARCHIVED | posts.view + posts.edit + posts.archive | Cancel schedule and retain private archived record. |
| WF-012 | Publish Archived | ARCHIVED | PUBLISHED | posts.view + posts.edit + posts.publish | Republish eligible archived content now. |
| WF-013 | Schedule Archived | ARCHIVED | SCHEDULED | posts.view + posts.edit + posts.schedule | Create future republish schedule. |
| WF-014 | Trash | DRAFT\|SCHEDULED\| PUBLISHED\|ARCHIVED | TRASH | posts.view + posts.edit + posts.trash | Clear active schedule/current public eligibility; preserve recoverable record. |
| WF-015 | Restore | TRASH | DRAFT | posts.view + posts.edit + posts.restore | Always restore privately; no automatic republish/reschedule. |

| WF ID | Action | From | To | Authorization reference | Semantic result |
|---|---|---|---|---|---|
| WF-016 | Permanent Delete | TRASH | TERMINAL | posts.view + posts.edit + posts.delete | Only eligible trashed record; retention/security policy may still block. |
| WF-017 | Edit Scheduled | SCHEDULED | SCHEDULED | posts.view + posts.edit | Commit content changes preserving schedule if still valid. |
| WF-018 | Edit Archived | ARCHIVED | ARCHIVED | posts.view + posts.edit | Commit private archived changes. |
| WF-019 | Revision Restore | DRAFT\|SCHEDULED\|PUBLISHED\|ARCHIVED | SAME STATUS | posts.view + posts.edit + posts.revisions.view + posts.revisions.restore | Restore eligible historical content fields into a new current version/revision; preserve post/site identity, lifecycle status, active schedule, publication timestamps and later revisions. Re-apply Section 5.2 field-level authorization to every field/relationship that would change; if any required field permission is missing, reject the entire restore atomically. Current validation and URL-history rules apply. |

**NOT IN V1 LIFECYCLE**
There is no Pending Review/Approval state, Scheduled Unpublish state, PUBLISHED -> DRAFT shortcut, cross-site Move state or hidden auto-publish-on-restore behavior. Adding any such state/transition changes feature semantics and requires reopening upstream scope.

## 15. Draft Creation and Durable Editing

- Create Post (ADM-S-POST-02) creates a durable DRAFT owned by current trusted SiteContext. site_id is server-derived.
- Initial create may include approved article content/relationships under create-mode authorization; after identity exists, further mutations use edit authorization.
- Save Draft is valid only while DRAFT and commits the working copy as current canonical content, increments version and creates a formal revision.
- Editing SCHEDULED or ARCHIVED uses the generic Edit Post capability and commits under the same status; it is not mislabeled as Save Draft.
- Editing PUBLISHED does not change public canonical content until Update Published succeeds; autosave remains private.

## 16. Scheduling, Rescheduling and Unscheduling

Scheduling uses the site timezone for editor intent but persists a single unambiguous absolute instant so later timezone configuration changes do not silently move an already-approved publication.

| Concern | Locked rule |
|---|---|
| Editor input | Local date/time + current site timezone identifier. |
| Validation | Resolve to exactly one future instant. Ambiguous/non-existent local times are rejected for explicit correction; the system must not guess. |

| Concern | Locked rule |
|---|---|
| Persisted schedule | Store absolute scheduled_for_utc plus the original local input/timezone context for traceability. |
| Timezone change later | Existing scheduled_for_utc remains unchanged. The calendar may display it in the new site timezone, but changing intended publication time requires explicit Reschedule. |
| Reschedule | Creates new durable Post version/revision and a new schedule identity/generation. Prior pending execution becomes stale. |
| Remove schedule | SCHEDULED -> DRAFT; clears active schedule; creates durable revision. |
| Worker execution | Reload by site_id + post_id; require site ACTIVE, status SCHEDULED, current schedule/version identity unchanged, due instant reached and publish validation still passes. |
| Stale/retried job | No duplicate publish. A stale generation/version is safe no-op/failure according to later worker contract. |

# 17. Publish and Update Published

## 17.1 Publish

- Publish requires a legal DRAFT/SCHEDULED/ARCHIVED source and all hard validation blockers to pass.
- Transition sets status PUBLISHED, clears active schedule, sets first_published_at if never set, sets published_at for this publication episode, increments version and creates revision.
- Successful Publish changes ordinary public eligibility; later systems update cache/search/sitemap/SEO artifacts through defined downstream effects.
- A scheduled Post manually published early invalidates the pending schedule so worker retry cannot publish it again.

## 17.2 Update Published

- PUBLISHED remains PUBLISHED.
- Editor/autosave working content is private until explicit Update Published commits it.
- Successful update increments version, creates revision and updates updated_at, but does not reset first_published_at or start a new publication episode timestamp.
- Failure/conflict leaves existing public version unchanged.

# 18. Archive Behavior

- Archive is legal from PUBLISHED and SCHEDULED; DRAFT does not require an Archive transition.
- PUBLISHED -> ARCHIVED removes ordinary public eligibility while retaining the Post, revisions, URL history and relationships.
- SCHEDULED -> ARCHIVED cancels the active schedule before archive commit.
- ARCHIVED may be edited privately and later explicitly Published or Scheduled using their own permissions.
- Archive does not destroy historical attribution, public URL history or revisions.

# 19. Trash, Restore and Permanent Delete

| Action | Precondition | Result | Important safety rule |
|---|---|---|---|
| Trash | DRAFT/SCHEDULED/PUBLISHED/ ARCHIVED + exact permission | TRASH | Clear active schedule; remove public eligibility; retain recoverable record/history. |
| Restore | TRASH + posts.restore path | DRAFT | Never return directly to PUBLISHED/SCHEDULED; explicit action required afterwards. |
| Permanent Delete | TRASH + posts.delete path + retention eligibility | Terminal removal | Exact DB retention/legal/security mechanics remain PDFs 08/19. Permanent Delete is terminal and does not create a new Post revision; deletion accountability is handled by the audit/security authority. No direct delete from a non-TRASH state. |

If a previously public Post is trashed and later restored, its existing URL history remains attached. The restored DRAFT is private until an explicit later publish/schedule succeeds.

# 20. Duplicate Post Semantics

Duplicate creates a new same-site DRAFT with independent identity/history. It is a content-copy convenience, not cloning of public state, schedules, history or media binaries.

| Concern | Locked duplicate rule |
|---|---|
| Eligible source | DRAFT, SCHEDULED, PUBLISHED or ARCHIVED. TRASH is not duplicable through normal Duplicate. |
| New identity | New post_id under same site_id; new version sequence; new initial revision. |
| Copied | Current committed title, excerpt, body, same-site author/taxonomy refs, featured/body media refs, manual related refs and approved SEO field values where safe. |
| Cleared/reset | status -> DRAFT; schedule fields; first/published/archived/trashed timestamps; URL history; revision history; autosaves. |
| Slug | Cleared so the new Post must obtain its own site-scoped slug before public eligibility. |
| Canonical override | Cleared because it can intentionally bind to the source public locator; editor may set a new authorized value later. |
| Media | Existing same-site media references may be reused; binaries/assets are not duplicated by Post Duplicate. |
| Authorization | posts.view + posts.create + posts.duplicate, plus normal same-site/business validation. |

# 21. Autosave and Recovery

Autosave exists to prevent loss of in-progress editor work. It is deliberately not a durable editorial version and must never cause public changes.

| Autosave concern | Locked behavior |
|---|---|
| Identity | Site + post + Admin account + editor instance/tab identity. This prevents one tab/site from overwriting another working session. |
| Payload | Recoverable editable working fields/body plus base_version; excludes publish/schedule/archive/trash/restore commands. |
| Lifecycle effect | None. Autosave never changes status, public eligibility, URL history, revision sequence or current canonical Post version. |
| Published editing | May hold unpublished working changes while current PUBLISHED version remains public. |
| Recovery detection | When a newer recoverable autosave exists, the UI may offer recovery; exact dialog/flow is PDF 10. |
| Accept recovery | Loads working copy only. Explicit Save Draft/Update Published/etc. is still required for durable commit. |
| After durable save | Corresponding autosave may be marked consumed/stale or cleared; exact storage retention is PDF 08/19 authority. |
| Site switching | Autosave is keyed by site; it must never appear in another SiteContext. |

**AUTOSAVE != REVISION**
A timer-based autosave must not create formal revisions, increment the Post version, trigger search/cache/sitemap updates or imply successful publication.

# 22. Formal Revisions and Revision Restore

## 22.1 Revision record semantic contract

| Revision property | Semantic meaning |
|---|---|
| revision_id | Stable revision identity. |

| Revision property | Semantic meaning |
|---|---|
| site_id | Same site as owning Post. |
| post_id | Owning Post identity. |
| revision_number | Monotonic sequence per Post. |
| source_action | Create/Save/Schedule/Publish/Update/Archive/Trash/Restore/ RevisionRestore/BulkEdit/system transition as applicable. |
| actor | Admin identity or explicit SYSTEM actor. |
| resulting_post_version | Post version produced by this durable mutation. |
| created_at | Revision commit time. |
| snapshot | Immutable committed content snapshot plus workflow metadata needed to understand the change. |

### 22.2 Revision creation events

- Initial draft creation.
- Explicit durable Save Draft or committed edit of SCHEDULED/ARCHIVED.
- Schedule/reschedule/remove schedule.
- Publish or Update Published.
- Archive, Trash and Restore.
- Restore Revision.
- Durable bulk content/taxonomy/authorship mutations applied to a Post.
- Permanent Delete is not a revision-creation event. It is a terminal destructive operation; no new Post revision is created immediately before or as part of deletion. Exact record removal/retention is PDF 08 authority and deletion accountability/audit behavior is PDF 19 authority.

### 22.3 What Revision Restore restores

Revision Restore requires posts.view + posts.edit + posts.revisions.view + posts.revisions.restore under the current trusted SiteContext. It creates a new current committed version from an old snapshot after validating it against current rules. Later revisions are preserved. The restore operation never rewinds identity, lifecycle status, active schedule, publication timestamps, workflow authority or history itself.

| Restorable from snapshot | Never restored from old snapshot |
|---|---|
| title, excerpt, body | site_id / post_id |
| author/category/tag assignments when still valid same-site and the corresponding Section 5.2 field permissions are satisfied | current status |
| featured/body media refs when still valid same-site and media.view is satisfied | active schedule or schedule generation |
| manual related refs when still valid same-site | first_published_at / published_at / archive/trash timestamps |
| slug and approved SEO content fields only when seo.view + seo.edit and any other applicable Section 5.2 permissions are satisfied | current version / revision IDs / revision sequence / audit history |

**FIELD-LEVEL RESTORE RULE.** The four revision permissions authorize the Revision Restore operation itself. Before commit, the server compares the selected historical snapshot with the current committed Post and re-applies the field-level authorization rules in Section 5.2 to every field or relationship that would change. If any required field permission is missing, the entire restore is rejected atomically; no unauthorized field is silently skipped and no partial restore is committed.

If Revision Restore changes the current slug of a Post that has been publicly addressable, Section 23 URL-history semantics apply. Revision Restore is not allowed while TRASH; Restore the Post to DRAFT first.

## 23. Slug Change and URL History

DOC-07 owns when URL history must be recorded. PDF 09 owns the final URL format, slug normalization and routing/redirect mechanics.

| Situation | Locked behavior |
|---|---|
| Draft never publicly addressable | Slug edits do not require public URL-history creation because no public locator has existed yet. |
| Post has been publicly addressable | A successful slug/locator-input change records the previous public locator identity before the new value becomes current. |
| Archived/trashed previously-public Post | Historical public locator records remain; later private state does not erase them. |
| History record | At minimum identifies site_id, post_id, prior slug/locator identity, change time, actor/source revision and whether it represented an actual public address. Exact persistence is PDF 08. |
| Route dependency | History must be capable of preserving the prior canonical public locator identity supplied by PDF 09; it must not reconstruct an old address from current taxonomy only. |
| Redirect creation | Separate capability. URL history is recorded regardless; creating/managing a redirect additionally requires the frozen redirects authorization path. |
| Editor permissions | Changing Post slug uses seo.view + seo.edit. posts.edit alone does not authorize slug mutation. |
| History deletion | Normal Post Edit cannot delete URL history. Retention/administrative mechanics are later authority. |

# 24. Optimistic Concurrency and Conflict Handling

Every durable Post mutation, including Permanent Delete, is compare-and-swap against the current version. Silent last-write-wins is prohibited. Revision-producing mutations create a new formal revision; Permanent Delete is the terminal exception and creates no new Post revision.

| Step | Required behavior |
|---|---|
| 1 | Client/editor loads Post version N and any working autosave records base_version=N. |
| 2 | Durable mutation submits expected_version=N. |
| 3 | Server loads target inside SiteContext and checks current version. |
| 4 | If current==expected, validate authorization + business rules and commit atomically. Revision-producing mutations update the Post, increment version and create a revision. Permanent Delete performs the authorized terminal removal without creating a new Post revision. Record all required downstream/audit side effects. |
| 5 | If current!=expected, reject conflict. Do not overwrite, auto-merge or partially commit. |
| 6 | Preserve/recover the user working copy/autosave so the editor can compare/reload/reapply. Exact conflict UI belongs to PDF 10. |

## 24.1 Operations covered

- Save Draft / committed edit
- Schedule/reschedule/unschedule
- Publish / Update Published
- Archive / Trash / Restore
- Revision Restore
- Durable bulk changes per target
- Worker scheduled publish using current schedule/version identity

**NO SILENT MERGE**
Automatic text/block merging after a version mismatch is not part of v1. A future merge strategy would require explicit content-schema/concurrency approval and QA.

# 25. Bulk Operations and Per-Object Lifecycle Enforcement

Bulk actions reuse single-Post rules; there is no bulk-only lifecycle state or blanket authorization.

- First validate request SiteContext and that every selected ID belongs to that same site; wrong-site security failures are not treated as ordinary partial business failures.
- For each target, re-check action permission, expected/current version, current status and transition eligibility.
- After security/ownership validation, eligible business operations may complete per object and report partial success/failure as frozen in DOC-03.
- Each successful durable target increments its own version and creates its own revision.
- Bulk publish/archive/trash/taxonomy/author changes use the exact composite permissions from DOC-02; selection alone requires posts.view.

## 26. Downstream Effects and Worker Handoff

DOC-07 freezes when semantic effects are required but does not define queue names, retry syntax, cache keys, search index technology or sitemap generation format.

| Trigger | Required semantic effect |
|---|---|
| Publish / Scheduled Publish | Public eligibility becomes true; enqueue/trigger tenant-scoped cache invalidation/refresh, search indexing, sitemap/SEO updates and audit. |
| Update Published | Public version changes; downstream public/cache/search/SEO artifacts must reflect the new committed version after safe processing. |
| Archive / Trash | Public eligibility becomes false; remove/invalidate tenant-scoped public/search/sitemap/cache artifacts as applicable. |
| Restore to Draft | No public re-add; only editorial state changes. Public artifacts remain ineligible until explicit Publish. |
| Slug change on public history | URL-history/redirect protection semantics become applicable; exact routing/redirect process later. |
| Schedule / Reschedule | Create/replace tenant-aware background publication intent. Exact job payload/queue/idempotency is PDF 18. |
| Revision Restore on PUBLISHED | If Revision Restore changes a PUBLISHED Post, it is treated as a public-content mutation and triggers the same tenant-scoped cache, search, SEO/sitemap and other applicable public refresh effects as Update Published. If the slug changes, Section 23 URL-history rules also apply. |
| Bulk mutation of PUBLISHED Post | A successful bulk mutation of a PUBLISHED Post inherits the same downstream public refresh effects as the equivalent successful single-Post mutation, evaluated independently for each Post. |

## 27. Isolation, Authorization and Non-Disclosure Guardrails

| Guard ID | Release-blocking rule |
|---|---|
| ISO-CNT-01 | Never load/update a Post by post_id without site scope in ordinary Admin/public/worker paths. |
| ISO-CNT-02 | All relationship targets are same-site; client references do not grant cross-site authority. |
| ISO-CNT-03 | Preview grant/token is bound to site + target and cannot be replayed on another host/site. |
| ISO-CNT-04 | Autosaves/revisions/URL history carry site identity and cannot be queried tenant-blind. |
| ISO-CNT-05 | Duplicate stays same-site; no cross-site destination parameter exists in v1. |
| ISO-CNT-06 | Worker scheduled publish carries persisted site identity and revalidates state/ownership/lifecycle. |
| ISO-CNT-07 | Published public delivery is resolved from CTX-PUBLIC-SITE and never by loading all sites then filtering. |
| ISO-CNT-08 | Wrong-site objects are unavailable/non-disclosed; response must not confirm their existence. |
| ISO-CNT-09 | A visible field/control never bypasses its exact DOC-02 authorization path. |
| ISO-CNT-10 | Site DISABLED/ARCHIVED/DELETION_REQUESTED lifecycle gates can block ordinary publishing/public effects under DOC-04. |

## 28. Developer Acceptance Criteria

| Acceptance ID | Release-blocking condition |
|---|---|
| ACC-CNT-001 | Exactly five persisted Post status values exist: DRAFT, SCHEDULED, PUBLISHED, ARCHIVED, TRASH. |
| ACC-CNT-002 | Preview is implemented as a private mode/page handoff, not a persisted Post status. |
| ACC-CNT-003 | Restore is an action/transition and always returns TRASH to DRAFT. |
| ACC-CNT-004 | No restore operation can republish or reschedule without separately authorized Publish/Schedule. |
| ACC-CNT-005 | Post site_id is immutable and server-derived from trusted SiteContext on create. |
| ACC-CNT-006 | All author/category/tag/media/related Post references are validated same-site before commit. |
| ACC-CNT-007 | Cross-site Duplicate is impossible through v1 Post Duplicate. |
| ACC-CNT-008 | Article body accepts only the 12 approved block types and rejects unknown/raw HTML/script/style blocks. |
| ACC-CNT-009 | Body HEADING accepts only levels 2, 3 and 4; H1 cannot be authored in body. |
| ACC-CNT-010 | CTA/callout/disclaimer payload contains content/semantic data only, not arbitrary visual source or CSS. |
| ACC-CNT-011 | EMBED accepts only approved provider/type input and never stores raw iframe/HTML from the editor. |
| ACC-CNT-012 | Every committed body validates content_schema_version and type payloads. |
| ACC-CNT-013 | Publish/Schedule rejects blank title, missing meaningful body, missing required author, invalid slug, invalid same-site refs or invalid block payload. |
| ACC-CNT-014 | Missing optional excerpt/custom SEO fields/featured media/tags/related Posts do not become hidden publish blockers. |
| ACC-CNT-015 | Draft, Scheduled, Archived and Trash are unavailable from ordinary PUB-ARTICLE-01. |
| ACC-CNT-016 | PUBLISHED is the only ordinary public Article eligibility state. |
| ACC-CNT-017 | Autosave never changes Post status, current version, revision sequence, URL history or public output. |
| ACC-CNT-018 | Autosave key includes site + post + Admin/editor-instance separation so tabs/sites cannot overwrite each other. |
| ACC-CNT-019 | Recovering an autosave loads a working copy; explicit durable save is still required. |
| ACC-CNT-020 | Every durable mutation uses expected_version and rejects stale versions without silent overwrite. |
| ACC-CNT-021 | A conflict leaves the current committed/public version unchanged and preserves the submitter working copy/recovery path. |
| ACC-CNT-022 | Initial draft create and each successful revision-producing durable editorial/content mutation creates a formal revision; Permanent Delete is the terminal exception and creates no new Post revision. |

| Acceptance ID | Release-blocking condition |
|---|---|
| ACC-CNT-023 | Revision Restore requires the base capability gate posts.view + posts.edit + posts.revisions.view + posts.revisions.restore, then re-applies current Section 5.2 field-level authorization to every field/relationship that would change. If any required field permission is missing, the entire restore is rejected atomically; no unauthorized field is skipped. On success it creates a new current revision, preserves current lifecycle status and never deletes later history. If the current status is PUBLISHED and committed content changes, the restore triggers the same tenant-scoped public refresh effects as Update Published; if the slug changes, URL-history rules also apply. |
| ACC-CNT-024 | Revision Restore does not rewind post/site identity, status, active schedule, version counters or publication timestamps. |
| ACC-CNT-025 | Revision Restore while TRASH is blocked until Post Restore returns it to DRAFT. |
| ACC-CNT-026 | Schedule stores one absolute instant plus original local/timezone context. |
| ACC-CNT-027 | A later site timezone change does not silently move an existing scheduled instant. |
| ACC-CNT-028 | Ambiguous/non-existent local schedule times are rejected instead of guessed. |
| ACC-CNT-029 | Reschedule invalidates the earlier schedule generation so stale worker execution cannot publish. |
| ACC-CNT-030 | Scheduled worker revalidates site ACTIVE, site ownership, status SCHEDULED, schedule/version identity, due time and publish eligibility. |
| ACC-CNT-031 | Manual early publish from SCHEDULED invalidates the pending schedule. |
| ACC-CNT-032 | Update Published does not expose autosaved/working changes until explicit commit succeeds. |
| ACC-CNT-033 | Update Published keeps first_published_at stable and does not start a new publication episode timestamp. |
| ACC-CNT-034 | Archive from PUBLISHED removes ordinary public eligibility while retaining editorial history. |
| ACC-CNT-035 | Archive from SCHEDULED invalidates the active schedule. |
| ACC-CNT-036 | Trash from any eligible non-TRASH state removes active schedule/public eligibility and retains recovery history. |
| ACC-CNT-037 | Permanent Delete is accepted only from TRASH, uses the current expected_version, creates no new Post revision and may still be blocked by later retention/security rules; deletion accountability is handled by audit/security authority. |
| ACC-CNT-038 | Duplicate produces a new same-site DRAFT with new identity/history and no copied schedule/public lifecycle timestamps. |
| ACC-CNT-039 | Duplicate clears slug and canonical_override rather than unintentionally cloning the source public locator. |
| ACC-CNT-040 | Duplicate reuses same-site media references rather than duplicating media binaries. |
| ACC-CNT-041 | URL history is created when a previously publicly addressable Post changes locator input; never-public Draft slug edits need not create public history. |
| ACC-CNT-042 | URL history survives Archive/Trash/Restore and normal Post Edit cannot erase it. |
| ACC-CNT-043 | URL history records can preserve prior public locator identity rather than reconstructing old URLs from current taxonomy only. |
| ACC-CNT-044 | Creating a protective redirect is separately authorized; slug change does not silently grant redirect permission. |

| Acceptance ID | Release-blocking condition |
|---|---|
| ACC-CNT-045 | Post slug/SEO fields inside Post Edit still require the exact SEO permissions from DOC-02. |
| ACC-CNT-046 | Author reassignment requires posts.view + posts.edit + authors.view; authors.edit alone is insufficient. |
| ACC-CNT-047 | Existing media selection requires media.view and uploading inside the editor requires media.upload in addition to the owning edit flow. |
| ACC-CNT-048 | Bulk security/site validation is fail-closed; business eligibility may then report per-object partial results. |
| ACC-CNT-049 | Every successful bulk durable Post mutation increments that Post version and creates its own revision. A successfully changed PUBLISHED Post independently inherits the same downstream public refresh effects as the equivalent successful single-Post mutation. |
| ACC-CNT-050 | No Pending Review, Scheduled Unpublish, generic Pages model or page-builder state is introduced by DOC-07. |
| ACC-CNT-051 | All lifecycle/public transitions preserve the 14-site isolation rules and do not create site-to-site data movement. |
| ACC-CNT-052 | DOC-05 PUB-SYS-PREVIEW-01 and DOC-06 ADM-S-POST-01..04 handoffs remain intact. |

## 29. Traceability Requirements

| Requirement ID | Locked requirement | Upstream | Primary downstream |
|---|---|---|---|
| CNT-REQ-001 | Preserve exactly one immutable site owner per Post. | DOC01/04 | PDF08/15/21 |
| CNT-REQ-002 | Use only the five persisted lifecycle status values defined here. | DOC03 | PDF08/10/15/21 |
| CNT-REQ-003 | Treat Preview as private non-status mode. | DOC03/04/05 | PDF10/11/15/19/21 |
| CNT-REQ-004 | Create Post only as DRAFT in trusted SiteContext. | DOC02/03/04/06 | PDF08/10/15/21 |
| CNT-REQ-005 | Keep Save Draft, Schedule, Publish, Archive, Trash, Restore, Delete permissions distinct. | DOC02 | PDF10/15/21 |
| CNT-REQ-006 | Restore TRASH to private DRAFT only. | DOC02/03 | PDF10/15/21 |
| CNT-REQ-007 | Implement ArticleBody as versioned ordered block document. | DOC03 | PDF08/12/13/15 |
| CNT-REQ-008 | Allow only approved structured block types; prohibit raw HTML/code escape hatch. | DOC03 | PDF12/15/19/21 |
| CNT-REQ-009 | Keep article H1 outside body and allow H2/H3/H4 only. | DOC03 | PDF12/21 |
| CNT-REQ-010 | Validate same-site author/taxonomy/media/related refs. | DOC04 | PDF08/15/21 |
| CNT-REQ-011 | Use one author attribution model; do not create co-author feature. | DOC03 | PDF08/13/15 |
| CNT-REQ-012 | Keep optional category/tag/featured media/related fields non-mandatory for publish unless later approved. | DOC03 | PDF10/21 |
| CNT-REQ-013 | Apply field-level authorization even on Post Editor. | DOC02/06 | PDF10/15/21 |
| CNT-REQ-014 | Use posts.edit + target view permissions for relationship assignment as applicable. | DOC02 | PDF10/15/21 |
| CNT-REQ-015 | Use seo.view + seo.edit for Post slug/SEO mutation. | DOC02/03 | PDF10/15/16/21 |
| CNT-REQ-016 | Represent structured-data inputs as validated source fields, not raw JSON-LD. | DOC03 | PDF15/16/19 |
| CNT-REQ-017 | Preserve manual related Post list as same-site authored data; automatic related remains derived conditional. | DOC03/04 | PDF13/16/18 |

| Requirement ID | Locked requirement | Upstream | Primary downstream |
|---|---|---|---|
| CNT-REQ-018 | Separate autosave from formal revision history. | DOC03 | PDF08/10/15/21 |
| CNT-REQ-019 | Autosave must never publish or increment canonical version/revision. | DOC03 | PDF08/10/15/21 |
| CNT-REQ-020 | Create a formal revision for each successful revision-producing durable editorial/content mutation; Permanent Delete is excluded as a terminal destructive operation. | DOC03 | PDF08/10/15/21 |
| CNT-REQ-021 | Authorize Revision Restore through the DOC-02 four-key base gate, re-apply Section 5.2 field-level permissions to every changed field/relationship, reject atomically if any are missing, create a new current revision and preserve later history; when a PUBLISHED Post changes, apply the same tenant-scoped public refresh effects as Update Published, with URL-history rules if the slug changes. | DOC02/03 | PDF08/10/15/16/18/21 |
| CNT-REQ-022 | Revision Restore does not restore lifecycle/status/identity/version history fields. | DOC03/04 | PDF08/10/15/21 |
| CNT-REQ-023 | Use site timezone input plus immutable scheduled instant semantics. | DOC03 | PDF08/10/15/18/21 |
| CNT-REQ-024 | Timezone configuration changes do not shift existing schedule intent silently. | DOC03 | PDF10/18/21 |
| CNT-REQ-025 | Scheduled worker revalidates SiteContext/lifecycle/state/version before publish. | DOC04 | PDF15/18/21 |
| CNT-REQ-026 | Use optimistic version token for every durable mutation. | DOC01/03 | PDF08/10/15/20/21 |
| CNT-REQ-027 | Reject version mismatch without silent last-write-wins/auto-merge. | DOC03 | PDF10/15/21 |
| CNT-REQ-028 | Preserve working copy/autosave on conflict for recovery. | DOC03 | PDF10/15/21 |
| CNT-REQ-029 | Duplicate remains same-site and produces new DRAFT identity/history. | DOC03/04 | PDF08/10/15/21 |
| CNT-REQ-030 | Duplicate clears schedule/public timestamps/slug/canonical override/history. | DOC03 | PDF08/10/15/21 |
| CNT-REQ-031 | Record URL history when previously public locator input changes. | DOC03 | PDF08/09/15/16/21 |
| CNT-REQ-032 | Keep URL history separate from redirect authorization. | DOC02/03 | PDF09/10/15/16 |
| CNT-REQ-033 | Use only PUBLISHED for ordinary PUB-ARTICLE-01 delivery. | DOC01/05 | PDF11/13/15/16/21 |
| CNT-REQ-034 | Keep Preview private/no-store/noindex and site-bound. | DOC04/05 | PDF11/15/16/19/21 |
| CNT-REQ-035 | Preserve ADM-S-POST-02/03/04 capability allocation. | DOC06 | PDF10/25 |
| CNT-REQ-036 | Do not create extra Admin pages for lifecycle actions. | DOC06 | PDF10/25 |
| CNT-REQ-037 | Archive/Trash/Publish trigger required tenant-aware downstream effects. | DOC03/04 | PDF16/18/21 |
| CNT-REQ-038 | Restore to Draft does not reactivate public artifacts. | DOC03/04 | PDF16/18/21 |

| Requirement ID | Locked requirement | Upstream | Primary downstream |
|---|---|---|---|
| CNT-REQ-039 | Bulk operations reuse per-Post authorization/state/version semantics; each successfully changed PUBLISHED Post inherits the downstream public refresh effects of the equivalent successful single-Post mutation. | DOC02/03 | PDF10/15/16/18/21 |
| CNT-REQ-040 | Wrong-site targets remain non-disclosed across content relationships/history/autosave/revisions. | DOC02/04 | PDF08/15/21 |

## 30. Dependencies and Downstream Authority Handoff

| Authority | Required relationship to BLOG-DOC-07 |
|---|---|
| BLOG-DOC-01 | Upstream product/topology/publication invariants; no scope expansion. |
| BLOG-DOC-02 | Upstream permission/authorization authority; DOC-07 never creates permission keys. |
| BLOG-DOC-03 | Upstream capability existence; DOC-07 refines exact lifecycle/content semantics only. |
| BLOG-DOC-04 | Upstream site ownership/isolation/context; every content/revision/autosave/worker operation remains tenant-scoped. |
| BLOG-DOC-05 | Upstream public Article/Preview page inventory; only PUBLISHED serves ordinary Article. |
| BLOG-DOC-06 | Upstream Admin Post page IDs and capability allocations. |
| PDF 08 | Implements Post/revision/autosave/URL-history schema, physical SQL types, constraints, RLS and retention. |
| PDF 09 | Implements slug normalization, site-scoped URL patterns, canonical locator/history interpretation and route behavior. |
| PDF 10 | Implements exact Admin flows/states/dialogs for save/schedule/publish/conflict/recovery/revision restore. |
| PDF 11 | Implements public Article/Preview/non-public state behavior. |
| PDF 12 | Defines rendering/component anatomy for approved article blocks without changing stored semantics. |
| PDF 13 | Defines safe public projections of PUBLISHED content. |
| PDF 15 | Makes mutations/validation/version preconditions executable in HTTP contracts. |
| PDF 16 | Owns exact SEO fallback/canonical/schema/search/sitemap/redirect algorithms. |
| PDF 17 | Owns media storage/variant/delivery details while preserving same-site references. |
| PDF 18 | Owns job/queue/cache syntax/idempotency for scheduling and downstream effects. |
| PDF 19 | Owns security/session/embed/link/file safety and retention-sensitive deletion controls. |
| PDF 21 | Executes release-blocking workflow/concurrency/isolation acceptance tests. |
| PDF 22 | Maps DOC-07 requirements through API/data/QA implementation. |

**CROSS-DOCUMENT AUDIT RESULT**

No genuine contradiction was found between this content/workflow model and the six project-frozen upstream authorities. DOC-07 preserves their product scope, 14-site ownership model, permission registry, capability catalog, public page inventory and Admin page inventory; it only resolves semantics explicitly delegated to PDF 07.

# Appendix A. Canonical Post Field Matrix

| Field ID | Field | Owner/mutation concern | Lifecycle/public note |
|---|---|---|---|
| PST-FLD-001 | post_id | System | Immutable across lifecycle/revision restore. |
| PST-FLD-002 | site_id | SiteContext | Immutable; cross-site move prohibited. |
| PST-FLD-003 | content_schema_version | Content schema | Validated on every commit/preview. |
| PST-FLD-004 | status | Lifecycle action permission | Only 5 persisted values. |
| PST-FLD-005 | version | Concurrency | Expected by every durable mutation. |
| PST-FLD-006 | title | Post edit | May be blank in early DRAFT; required before schedule/publish; owns H1. |
| PST-FLD-007 | excerpt | Post edit | Optional. |
| PST-FLD-008 | body | Post edit | Meaningful valid body required for schedule/publish. |
| PST-FLD-009 | author_site_profile_id | Post edit + authors.view | Required before schedule/publish. |
| PST-FLD-010 | category_ids | Post edit + categories.view | Optional; same-site. |
| PST-FLD-011 | tag_ids | Post edit + tags.view | Optional; same-site. |
| PST-FLD-012 | featured_media_id | Post edit + media.view | Optional; same-site. |
| PST-FLD-013 | manual_related_post_ids | Post edit + posts.view | Optional; same-site/no self. |
| PST-FLD-014 | slug | seo.view + seo.edit | Required before public eligibility; history trigger if previously public. |
| PST-FLD-015..024 | SEO inputs | SEO/media/post field owner per DOC-02 | Optional values; exact emission/fallback later. |
| PST-FLD-025..027 | schedule fields | posts.schedule action | Present only while SCHEDULED. |
| PST-FLD-028 | first_published_at | System lifecycle | Immutable once first publication succeeds. |
| PST-FLD-029 | published_at | System lifecycle | Set on transition into PUBLISHED; not reset by Update Published. |
| PST-FLD-030 | archived_at | System lifecycle | Current archive timestamp. |
| PST-FLD-031 | trashed_at | System lifecycle | Current trash timestamp. |
| PST-FLD-032..035 | audit timestamps/actors | System | Durable mutation provenance; exact audit schema later. |

# Appendix B. Article Block Payload Matrix

| Block | Required payload | Optional payload | Prohibited/important rule |
|---|---|---|---|
| PARAGRAPH | RichText content | - | No visual style controls. |
| HEADING | level 2\|3\|4 + RichText | - | H1 invalid. |
| LIST | style + >=1 item | - | No arbitrary nested block tree. |
| QUOTE | quote | attribution, source | No raw markup. |
| SEPARATOR | - | - | No author-controlled CSS/variant code. |
| MEDIA | same-site media_id | alt_override, caption | No cross-site media. |
| TABLE | rectangular rows | header_row | No formulas/scripts/nested blocks. |
| EMBED | approved provider + source | caption | No raw iframe/HTML. |
| CALLOUT | body | label | Frontend owns styling. |
| CTA | label + destination | supporting_text | No CSS/component source. |
| DISCLAIMER | body | label | Frontend owns styling. |
| SOURCES_REFERENCES | >=1 source entry | url/publisher/note per entry | Validated URLs; ordered entries. |

# Appendix C. Lifecycle Transition Matrix

| WF ID | From | Action | To | Authorization / executor |
|---|---|---|---|---|
| WF-001 | NONE | Create | DRAFT | posts.view + posts.create |
| WF-002 | DRAFT | Save Draft | DRAFT | posts.view + posts.edit |
| WF-003 | DRAFT | Schedule | SCHEDULED | posts.view + posts.edit + posts.schedule |
| WF-004 | SCHEDULED | Reschedule | SCHEDULED | posts.view + posts.edit + posts.schedule |
| WF-005 | SCHEDULED | Remove Schedule | DRAFT | posts.view + posts.edit + posts.schedule |
| WF-006 | DRAFT | Manual Publish | PUBLISHED | posts.view + posts.edit + posts.publish |
| WF-007 | SCHEDULED | Early/Manual Publish | PUBLISHED | posts.view + posts.edit + posts.publish |
| WF-008 | SCHEDULED | Scheduled Publish | PUBLISHED | SYSTEM worker under CTX-WORKER-SITE |
| WF-009 | PUBLISHED | Update Published | PUBLISHED | posts.view + posts.edit |
| WF-010 | PUBLISHED | Archive Published | ARCHIVED | posts.view + posts.edit + posts.archive |
| WF-011 | SCHEDULED | Archive Scheduled | ARCHIVED | posts.view + posts.edit + posts.archive |
| WF-012 | ARCHIVED | Publish Archived | PUBLISHED | posts.view + posts.edit + posts.publish |
| WF-013 | ARCHIVED | Schedule Archived | SCHEDULED | posts.view + posts.edit + posts.schedule |
| WF-014 | DRAFT\|SCHEDULED\| PUBLISHED\|ARCHIVED | Trash | TRASH | posts.view + posts.edit + posts.trash |
| WF-015 | TRASH | Restore | DRAFT | posts.view + posts.edit + posts.restore |
| WF-016 | TRASH | Permanent Delete | TERMINAL | posts.view + posts.edit + posts.delete |
| WF-017 | SCHEDULED | Edit Scheduled | SCHEDULED | posts.view + posts.edit |
| WF-018 | ARCHIVED | Edit Archived | ARCHIVED | posts.view + posts.edit |
| WF-019 | DRAFT\|SCHEDULED\| PUBLISHED\|ARCHIVED | Revision Restore | SAME STATUS | Base gate: posts.view + posts.edit + posts.revisions.view + posts.revisions.restore; commit also requires every applicable Section 5.2 field-level permission for fields/relationships changed by the selected snapshot. Missing field permission rejects the restore atomically. |

# Appendix D. One-Page Developer Execution Reference

| Question | Locked answer |
|---|---|
| How many persisted Post statuses? | 5: DRAFT, SCHEDULED, PUBLISHED, ARCHIVED, TRASH. |
| Is Preview a status? | No. Private mode via PUB-SYS-PREVIEW-01; no state change. |
| What does Restore do? | TRASH -> DRAFT only. Never republish/reschedule implicitly. |
| Ordinary public Article eligibility? | PUBLISHED only. |

| Question | Locked answer |
|---|---|
| Autosave vs Revision? | Autosave = private recovery snapshot. Revision = immutable history of durable commit. |
| Concurrency? | Every durable mutation supplies expected_version; mismatch rejects without overwrite/auto-merge. |
| Schedule timezone? | Enter in site timezone, persist one absolute instant + original timezone context; later timezone changes do not silently shift it. |
| Body model? | Versioned ordered block document, 12 approved block types, no raw HTML/page builder. |
| H1? | Post title/page template owns H1. Body supports H2/H3/H4 only. |
| Duplicate? | Same-site new DRAFT, new identity/history; no schedule/public timestamps/slug/canonical override/history copied. |
| Slug/URL history? | Previously public locator changes create history; redirect creation is separately authorized. |
| Author/taxonomy/media? | Same-site references only; relationship selection uses exact DOC-02 field/action permissions. Revision Restore re-applies those field permissions (including SEO/media rules) to every changed field and rejects the entire restore if any required permission is missing. |
| Published edits? | Working/autosave stays private until Update Published commits. |
| Permanent Delete? | TRASH only; uses expected_version; creates no new Post revision; exact retention/removal and audit/security behavior remain later authorities. |
| Can a developer add Review/Approval state? | No. Not in v1 frozen capability scope. |
| Which Admin pages? | ADM-S-POST-01 List, -02 Create, -03 Edit/Article Editor, -04 Revision History; Preview hands to public Preview page. |

**FINAL DEVELOPER LOCK**
Implement this content/lifecycle semantic model exactly. Physical schema, API paths, UI controls, queue names and visual rendering may vary only within their later authority documents; they may not redefine these state, block, revision, autosave, schedule, duplicate, URL-history or concurrency meanings.